

ON THE II_1 FACTORS OF FUCHSIAN GROUPS

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ABSTRACT. We show that von Neumann algebras of fundamental groups of closed orientable surfaces of genus $g \geq 2$ are free group factors on $2g - 1$ generators. The key technical ingredient involves a proof that the element $w = ABA^{-1}B^{-1}$ of the free group $\mathbb{F}_2 = \langle A, B \rangle$ is freely complemented in the group factor: $L(\mathbb{F}_2) = W^*(w) * W^*(v)$ for some Haar unitary $v \in L(\mathbb{F}_2)$ that is freely independent from w . Combined with previous results, we conclude that for an arbitrary finitely generated torsion-free non-elementary discrete subgroup $\Gamma \subset PSL_2(\mathbb{R})$, $L(\Gamma)$ is a free group factor, settling a conjecture of de la Harpe and Voiculescu. This result was obtained using OpenAI's ChatGPT Pro 6.0.

1. INTRODUCTION.

Recall (see e.g. [9]) that a finitely generated non-elementary (i.e., not finite and not a finite extension of $\mathbb{Z}$) discrete subgroup of $PSL_2(\mathbb{R})$ has the presentation

$$\Gamma(g : m_1, \dots, m_k : t, s) = \left\langle a_1, b_1, \dots, a_g, b_g, x_1, \dots, x_k, p_1, \dots, p_s, h_1, \dots, h_t \mid \right. \\ \left. \forall j, x_j^{m_j} = 1, \left(\prod_{i=1}^g a_i b_i a_i^{-1} b_i^{-1} \right) \cdot x_1 \cdots x_k \cdot p_1 \cdots p_s \cdot h_1 \cdots h_t = 1 \right\rangle$$

where $g, k, t, s \geq 0$, $m_1, \dots, m_k \geq 2$ are satisfy the condition

$$r := 2g - 1 + s + t + \sum_{j=1}^r \left(1 - \frac{1}{m_j} \right) > 1.$$

In [9, Problem on p. 157] de la Harpe and Voiculescu asked whether it's always true that $L(\Gamma(g : m_1, \dots, m_k, t, s)) \cong L(\mathbb{F}_r)$; moreover, they showed that this is true if $t + s > 0$. (If r is non-integer, $L(\mathbb{F}_r)$ stands for an interpolated free group factor [5, 17].)

If the group is torsion-free (i. e. $k = 0$) and $t + s > 0$, the the answer is immediate (as observed in [9]), since the group itself is a free group. The remaining torsion free case corresponds to $t = s = k = 0$ and $g \geq 2$. In that case, the group is naturally an amalgamated free product of two free groups, with amalgamation over an infinite cyclic subgroup:

$$\Gamma(g :: 0, 0) = \langle a_1, b_1 \rangle *_{a_1 b_1 a_1^{-1} b_1^{-1} = (\prod_{j=2}^g a_j b_j a_j^{-1} b_j^{-1})^{-1}} \langle a_2, b_2, \dots, a_g, b_g \rangle$$

and its isomorphism class remained open until now (see the discussion of the Problem on p. 157 of [9]).

The main technical result of this paper, Theorem 3, shows that there exists a Haar unitary v , for which $L(\mathbb{F}_2) = W^*(a_1, b_1) = W^*(a_1 b_1 a_1^{-1} b_1^{-1}) * W(v)$. In other words, the abelian

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von Neumann subalgebra of $L(\mathbb{F}_2)$ generated by the commutator $a_1 b_1 a_1^{-1} b_1^{-1}$ is freely complemented (see [1, Remark 1.4(2)] for a discussion of this question; see also [16, \S3.2] and [15]). This implies (see Corollary 3) that the amalgamated free product above is indeed a free group factor:

$$W^*(v) * W^*(a_2, b_2, \dots, a_g, b_g) \cong L(\mathbb{F}_{2g-1}).$$

The following theorem summarizes all known results for discrete finitely generated non-elementary subgroups of $PSL_2(\mathbb{R})$; cases 2 and 3 are new and are proved in this paper:

Theorem 1. *Suppose that any of the following conditions hold:*

- (1) $t + s > 0$
- (2) $g \geq 2$
- (3) $g \geq 1$ and $k \geq 2$

Then $L(\Gamma(g; m_1, \dots, m_k : t, s)) \cong L(\mathbb{F}_r)$ where $r = 2g - 1 + s + t + \sum_{j=1}^g (1 - \frac{1}{m_j})$.

Proof. The first case corresponds to cases (2) and (4) described on p. 156 of [9]. As noted there, the group $\Gamma(g : m_1, \dots, m_k : t, s)$ is a free product of cyclic groups, and $L(\Gamma(g; m_1, \dots, m_k : t, s)) \cong L(\mathbb{F}_r)$ by the results of Dykema [6].

In the second and third cases, consider the group K freely generated by $a_2, b_2, \dots, a_g, b_g$ as well as $x_1, \dots, x_k$ subject only to $x_j^{m_j} = e$ (if $g = 1$, we only have $x_1, \dots, x_k$ as generators). Thus

$$K = \mathbb{Z}/m_1\mathbb{Z} * \dots * \mathbb{Z}/m_k\mathbb{Z} * (\mathbb{Z})^{*(2g-2)}.$$

Put

$$z = \prod_{i=2}^g a_i b_i a_i^{-1} b_i^{-1} \cdot x_1 \dots x_k$$

Then the assumptions that $g \geq 2$ or $k \geq 2$ guarantee that z has infinite order.

Let $\Lambda = \mathbb{F}_2$ generated by two elements a_1 and b_1 . Then

$$\Gamma(g : m_1, \dots, m_k : 0, 0) \cong \Lambda *_{a_1 b_1 a_1^{-1} b_1^{-1} = z^{-1}} K.$$

As before, let $\mathcal{A} = W^*(aba^{-1}b^{-1}) \subset L(\Lambda)$. Applying Corollary 3

$$\begin{aligned} L(\Gamma(g : m_1, \dots, m_k : 0, 0)) &\cong L(\mathbb{F}_2) *_{\mathcal{A}} L(K) \\ &\cong L(\mathbb{Z}) * L(K) \cong L(\mathbb{Z} * K) \\ &\cong L(\mathbb{Z}/m_1\mathbb{Z} * \dots * \mathbb{Z}/m_k\mathbb{Z} * (\mathbb{Z})^{*(2g-1)}) \\ &\cong L(\Gamma(0 : m_1, \dots, m_k : 2g)) \cong L(\mathbb{F}_{2g-1+\sum(1-m_j^{-1})}) \end{aligned}$$

where the last isomorphism uses the first case of the Theorem. □

Corollary 1. *If Λ is a torsion-free finitely degenerated non-elementary discrete subgroup of $PSL_2(\mathbb{R})$, then $L(\Gamma) \cong L(\mathbb{F}_r)$ (where r is as above). In particular, if $\Lambda = \pi_1(\Sigma_g)$ is the fundamental group of a surface of genus $g \geq 2$, then $L(\Lambda) \cong L(\mathbb{F}_{2g-1})$.*

Proof. The torsion-free case corresponds to $k = 0$. Since $r = 1 + 2g - 2 + s + t > 1$ this means that either $s + t > 0$, or $g \geq 2$, corresponding to parts 1 or 2 of Theorem 1. □

While we are unable to settle the de la Harpe-Voiculescu conjecture in all cases, we are able to prove a weaker statement of containment in (or of) an interpolated free group factor up to finite index:

Theorem 2. *Let Γ be a finitely generated non-elementary discrete subgroup of $PSL_2(\mathbb{R})$. Then $L(\Gamma)$ contains a free group factor as a finite-index subfactor. Moreover, $L(\Gamma)$ is contained in an interpolated free group factor as a finite-index subfactor.*

Proof. The remaining families of discrete subgroups of $PSL_2(\mathbb{R})$ not covered by Theorem 1 are:

The cases $\Gamma(0 : m_1, \dots, m_k : 0, 0)$ with $m_1, \dots, m_k$ satisfying $\sum(1 - m_j^{-1}) > 2$ (thus necessarily $k \geq 3$). The group presentation is $\langle x_1, \dots, x_k \mid x_j^{m_j} = 1, x_1 \cdots x_k = 1 \rangle$. The prediction is that the group von Neumann algebra should be $L(\mathbb{F}_r)$ with $r = (\sum(1 - m_j^{-1})) - 1$.

The cases $\Gamma(1 : m : 0, 0)$. The presentation is $\langle a, b \mid (aba^{-1}b^{-1})^m = 1 \rangle$, for some $m \geq 2$. The prediction is that the von Neumann algebra should be $L(\mathbb{F}_r)$ with $r = 2 - 1/m$.

In either of these cases, Γ is a co-compact subgroup of $PSL_2(\mathbb{R})$. By the solution of the Fenchel's Conjecture (see [13, 2, 7, 3, 11, 12]), there exists a finite-index torsion-free subgroup $\Lambda \subset \Gamma$. Since $\Gamma \subset PSL_2(\mathbb{R})$ is co-compact, Λ is also co-compact and thus by Corollary 1, $L(\Lambda) \cong L(\mathbb{F}_k)$ for some integer k . Thus $L(\mathbb{F}_k) \cong L(\Lambda) \subset L(\Gamma)$ has finite index.

Using the Jones basic construction [10] we also obtain the inclusion $L(\Gamma) \subset \langle L(\Gamma), e_{L(\Lambda)} \rangle$. By [14], $\langle L(\Gamma), e_{L(\Lambda)} \rangle$ is the amplification $L(\Lambda)^{[\Gamma:\Lambda]}$. Using the amplification formula for free group factors [5, 6, 17] we conclude that M_1 is also an interpolated free group factor. $\square$

It is worth mentioning that our result can be viewed as a group von Neumann algebra counterpart to the work of Conley, Gaboriau, Marks and Tucker-Drob ([4], see also [8]) where a similar “free complementation” result was proved in the context of equivalence relations generated by group actions with planar Cayley graphs (in particular, fundamental groups of surfaces).

1.1. How this result was obtained: use of AI. Our original approach to the free complementation question (from some years ago) was to construct a free complement to w by finding a family of “changes of generators” that would transform A, B into the pair (w, v) and thus would yield a free complement v to w , so that $W^*(u, v)$ still generates all of $W^*(A, B)$. This approach is natural from free transportation theory. It is also natural to try to create such a path of automorphisms α_t by solving a differential equation, in effect, integrating a time-dependent divergence-free vector field. To keep the changes of variables invertible, one needs to also set up a differential equation for the inverse transformation, and to show that these differential equations have solutions that behave well. Since not every unitary in $L(\mathbb{F}_2)$ is freely complemented (or even generates a MASA) one needs to extract specific analytical features from the particular case of $w = ABA^{-1}B^{-1}$, and our initial starting point was certain combinatorial considerations connected to the proof of the ergodic theory version of the problem [4]. ChatGPT Pro 5.6 was then used to experiment with various approaches: showing existence of solutions, invertibility of changes of variables, etc., all aiming to obtain and propagate estimates involving the particular geometry of the subgroup generated by w . The manuscript was approaching 70 pages, but we were still unable to find the right estimates that would close the proof. We then asked ChatGPT Pro 6.0 to take a look at the manuscript and suggest improvements in the approach. After some thought, it returned a 7-page proof that became the basis of this manuscript. In its own words:

The argument arose from an attempt to obtain free complementation by integrating constrained time-dependent derivations, motivated by measured free-factor constructions of [4]. Although the proof presented here does not use

that transport construction, the elementary commutator-preserving automorphisms and the need to preserve generation in the limit emerged from its analysis. The resulting argument instead uses finite changes of generators, the associated tree geometry, and quantitative preservation of polynomial approximations.

This work was developed with substantial assistance from ChatGPT (OpenAI), used interactively to explore proof strategies, propose and critique intermediate arguments, carry out exploratory calculations, and draft and revise the manuscript. This assistance included mathematical content, in particular the development of the finite-automorphism approach and its Fourier estimates, and was not limited to editing. The author supplied the research question and initial geometric motivation, and directed the investigation.

The development proceeded through an extended sequence of proposals, objections, corrections, and changes of approach; the final argument should not be understood as an unmodified output from a single prompt.

The argument presented by ChatGPT was then carefully checked and rewritten by the author. The author is fully responsible for the content of this manuscript, including the mathematical claims, proofs, and references.

1.2. Outline of the construction. The free complement v is obtained as a limit of elements v_n , which are themselves constructed as $v_n = (\beta_1 \circ \cdots \circ \beta_n)(A)$, where $\beta_1, \dots, \beta_n$ are certain automorphisms of $L(\mathbb{F}_2)$ that satisfy $\beta_j(w) = w$. Since A and w are freely independent, it follows that v_n and w are free at all times. The automorphisms β_j are chosen in such a way that v_n form a Cauchy sequence, so that $v_n \rightarrow v$. What is key, however, is to ensure that $W^*(v, w) = L(\mathbb{F}_2)$. This is guaranteed by choosing automorphisms in such a way that both A and B are well-approximated by v_n and w . Writing $N = W^*(A, w)$, the approximation is achieved by choosing Θ_n so that the conjugated conditional expectation $E_{\Theta_n(N)} = \Theta_n \circ E_N \circ \Theta_n^{-1}$ approximates the identity on prescribed finite sets. Proposition 1 formalizes the passage from such local approximation to free complementation through a Baire category argument. In proposition 2 we show that instead of controlling $\Theta_n \circ E_{W^*(A, w)} \circ \Theta_n^{-1}$, one can instead use a family of positive L^2 contractions $\mathbf{T}_k$ interpolating between $E_{W^*(A, w)}$ and identity, and prove a certain inequality involving the quadratic form $\mathbf{Q}_k$ associated to $\mathbf{T}_k$.

The remainder of the paper is dedicated to producing a family of automorphisms that satisfy the hypothesis of Proposition 2 and the choice of the family $\mathbf{T}_k$. The automorphisms β_j are obtained by repeatedly composing automorphism of the form

$$R_{-m} \circ S_{f_{m,t}} \circ R_m$$

where $R_m(A) = A$, $R_m(B) = BA^m$, and $S_f(B) = B$, $S_{f_{m,t}}(A) = Af(B)$ for certain functions $f_{m,t} : \mathbb{T} \rightarrow \mathbb{T}$. The contractions $\mathbf{T}_k$ are diagonal in the group basis, and multiply an element h by $q_k^{\ell(h)}$ where $\ell(h)$ is the length of h relative to the subgroup generated by w and A , and $q_k \uparrow 1$, $q_0 = 0$ are carefully chosen, along with a family of functions $f_{m,t}$. The key estimate is given by Lemma 9.

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2. FREE COMPLEMENTATION.

Let $\mathbb{F}_2 = \langle A, B \rangle$ be the free group on two generators, and let $M = L(\mathbb{F}_2)$. Regard $\mathbb{F}_2 \subset M$. We endow M with its unique normalized group trace τ .

Suppose that $w \in M$ is a Haar unitary so that w is free from A , and put $N = W^*(w, A) \subset M$. We give the following sufficient condition for the existence of an abelian free complement to w , i.e., the existence of a Haar unitary v free from w and so that $W^*(w, v) = M$:

Proposition 1. *Suppose that for any $x_1, \dots, x_n \in M$ and any $\epsilon > 0$, $\eta > 0$, there exists an automorphism $\theta : M \rightarrow M$ so that*

- $\theta(w) = w$
- $\|\theta(A) - A\|_2 < \eta$
- $\sum_{j=1}^n \|x_j - E_{\theta(N)}(x_j)\|_2^2 < \epsilon$.

Then for any $\delta > 0$ there exists a Haar unitary $v \in M$ so that:

- v is free from w ,
- $M = W^*(w, v) = W^*(w) * W^*(v)$
- $\|v - A\|_2 < \delta$.

In particular, $W^(w)$ is freely complemented in M by the abelian algebra $W^*(v)$.*

Proof. Let $X = \overline{\{\theta(A) : \theta \in \text{Aut}(M), \theta(w) = w\}}^{L^2(M)}$. Since the unitary group of M is closed in the L^2 norm and X is a closed subset, X is complete. Since A is a Haar unitary free from w , any automorphism fixing w takes A to a Haar unitary free from w . Since freeness is preserved under limits, it follows that every element of X is a Haar unitary free from w .

Now choose a total sequence $x_k \in M$ for $\|\cdot\|_2$. For each $n > 0$ let

$$O_n = \{u \in X : \exists \text{ *-polynomials } P_1, \dots, P_n \text{ with } \max_{1 \leq j \leq n} \|x_j - P_j(w, u)\|_2 < 1/n\}.$$

Since each P_j is Lipschitz in its unitary arguments for the L^2 norm, the set O_n is open. Furthermore, O_n is dense. Indeed, suppose that $\theta_0 \in \text{Aut}(M)$, $\theta_0(w) = w$, and $\rho > 0$ are given. Then we can find an automorphism θ fixing w and such that $\|\theta(A) - A\|_2 < \rho$ and

$$\sum_{j=1}^n \|\theta_0^{-1}(x_j) - E_{\theta(N)}(\theta_0^{-1}(x_j))\|_2^2 < 1/(4n^2).$$

Equivalently,

$$\sum_{j=1}^n \|x_j - E_{\theta_0(\theta(N))}(x_j)\|_2^2 < 1/(4n^2).$$

Thus x_j is approximated by elements $E_{\theta_0(\theta(N))}(x_j)$ of $\theta_0(\theta(N)) = W^*(\theta_0(\theta(A)), w)$ to within $1/(2n)$ in L^2 . Therefore, we can find *-polynomials $P_1, \dots, P_n$ with the property that $P_j(w, \theta_0(\theta(A)))$ are within $1/2n$ of $E_{\theta_0(\theta(N))}(x_j)$ and so within $1/n$ of x_j . Thus $(\theta_0 \circ \theta)(A) \in$

O_n . On the other hand, $\|\theta_0(\theta(A)) - \theta_0(A)\|_2 < \rho$ so that $\theta_0(A)$ is within ρ of O_n . Since the set $\{\theta(A) : \theta \in \text{Aut}(M), \theta(w) = w\}$ is dense in X , it follows that O_n is dense as well.

By Baire Category Theorem, the intersection of the sets O_n is dense. Choose a unitary v in this intersection so that $\|v - A\|_2 < \delta$. Then v is free from w , and, since for each j , $v \in O_n$ for arbitrary large $n > j$, it follows that $x_j \in L^2(W^*(w, v))$. Since $\{x_j\}_j$ is total, it follows that $W^*(w, v) = M$. $\square$

Proposition 2. *Suppose that there exist positive contractions $\mathbf{T}_k : L^2(M) \rightarrow L^2(M)$, $k \geq 0$, so that $\mathbf{T}_0 = E_N$ and $\mathbf{T}_k \rightarrow \text{Id}$ strongly. Put $\mathbf{Q}_j(x) = \langle \mathbf{T}_j x, x \rangle$.*

Suppose moreover that for any $k, n \geq 0$ and any $\epsilon > 0, \eta > 0$, and any $x_1, \dots, x_n \in M$, there exists an automorphism $\theta : M \rightarrow M$ so that:

- $\theta(w) = w$
- $\|\theta(A) - A\|_2 < \eta$
- $\sum_{i=1}^n \mathbf{Q}_k(\theta^{-1}(x_i)) \geq \sum_{i=1}^n \mathbf{Q}_{k+1}(x_i) - \epsilon$.

Then the hypothesis of Proposition 1 is satisfied, and thus its conclusion holds.

Proof. Let $x_1, \dots, x_n, \eta, \epsilon$ be given as in the hypothesis of Proposition 1. Choose $J \geq 1$ large enough so that

$$\sum_i \mathbf{Q}_J(x_i) > \sum_i \|x_i\|_2^2 - \epsilon/2.$$

Let $x_i^{(J)} = x_i$, and construct $x_i^{(J-1)}, \dots, x_i^{(0)}$ recursively as follows. Having chosen $x_i^{(j)}$ and automorphisms $\theta_{j-1}, \dots, \theta_j$, choose θ_{j-1} so that $\theta_{j-1}(w) = w$, $\|\theta_{j-1}(A) - A\|_2 < \eta/2J$,

$$\sum_i \mathbf{Q}_{j-1}(\theta_{j-1}^{-1}(x_i^{(j)})) \geq \sum_i \mathbf{Q}_j(x_i^{(j)}) - \epsilon/2J,$$

and set $x_i^{(j-1)} = \theta_{j-1}^{-1}(x_i^{(j)})$. Then

$$\sum_i \mathbf{Q}_{j-1}(x_i^{(j-1)}) \geq \sum_i \mathbf{Q}_j(x_i^{(j)}) - \epsilon/2J.$$

Let now $\Theta = \theta_{J-1} \circ \dots \circ \theta_0$. Since $\mathbf{T}_0 = E_N$ is an orthogonal projection, and noting that $\Theta^{-1}(x_i) = x_i^{(0)}$, we get that $\|\Theta(A) - A\|_2 < \eta$ and

$$\begin{aligned} \sum_i \|x_i - E_{\Theta(N)}(x_i)\|_2^2 &= \sum_i \|\Theta^{-1}(x_i) - E_N(\Theta^{-1}(x_i))\|_2^2 \\ &= \sum_i \left[\|x_i\|_2^2 - \langle E_N(x_i^{(0)}), x_i^{(0)} \rangle \right] \\ &= \sum_i \left[\|x_i\|_2^2 - \mathbf{Q}_0(x_i^{(0)}) \right] \\ &\leq \sum_i \left[\|x_i\|_2^2 - \mathbf{Q}_1(x_i^{(1)}) \right] + \epsilon/2J \\ &\leq \dots \leq \sum_i \left[\|x_i\|_2^2 - \mathbf{Q}_J(x_i^{(J)}) \right] + \epsilon/2 \\ &< \sum_i \left[\|x_i\|_2^2 - \|x_i\|_2^2 \right] + \epsilon/2 + \epsilon/2 = \epsilon, \end{aligned}$$

Similarly, $\|\Theta(A) - A\|_2 \leq \sum_{j=0}^{J-1} \|\theta_j(A) - A\|_2 \leq \eta/2 < \eta$. $\square$

3. THE CASE OF THE COMMUTATOR MASA.

Let as before $M = L(\mathbb{F}_2)$ with group generators A and B , and let $w = ABA^{-1}B^{-1}$ be their group commutator. Note that w and A are free inside $\mathbb{F}_2$. Let H denote the subgroup generated by A and w . Set $N = W^*(A, w) = L(H)$.

3.1. A class of automorphisms. We first describe a class of automorphisms of M that will be used to deduce the hypothesis of Proposition 2.

Lemma 1. *Let $f : \mathbb{T} \rightarrow \mathbb{T}$ be a measurable function. Then the map $A \mapsto Af(B)$, $B \mapsto B$ extends to a normal trace-preserving automorphism $S_f : M \rightarrow M$. The inverse of S_f is $S_{\bar{f}}$, and $S_f(w) = w$.*

Proof. Since A and B are two free Haar unitaries, it's immediate to verify that also $Af(B)$ and B are free Haar unitaries. Thus the map $A \mapsto Af(B)$, $B \mapsto B$ extends to a normal injective $*$ -homomorphism S_f from M to itself. Since $B \in W^*(Af(B), B)$, also $A = (Af(B))\bar{f}(B) \in W^*(Af(B), B)$ and thus $W^*(Af(B), B) = M$. Hence S_f is onto and thus an automorphism. Clearly, $S_{\bar{f}}S_f(B) = B$ and $S_{\bar{f}}S_f(A) = S_{\bar{f}}(Af(B)) = A\bar{f}(B)f(B) = A$, so $S_{\bar{f}}$ is the inverse to S_f . Finally, $S_f(w) = S_f(ABA^{-1}B^{-1}) = Af(B)Bf(B)^{-1}A^{-1}B^{-1} = w$. $\square$

Lemma 2. *For each $m \in \mathbb{Z}$, the map $A \mapsto A$, $B \mapsto BA^m$ extends to a normal trace-preserving automorphism $R_m : M \rightarrow M$. The inverse of R_m is R_{-m} , and $R_m(w) = w$.*

Proof. The automorphism R_m is conjugate to S_{z^m} via the map that exchanges A and B . Moreover, $R_m(w) = ABA^mA^{-1}A^{-m}B^{-1} = w$. $\square$

It will be useful to give a more detailed description of the automorphism S_f . To this end, suppose that $g \in \mathbb{F}_2$ is fixed. Suppose that g , as a reduced word in generators A and B , contains a total of $p \geq 1$ occurrences of $A^{\pm 1}$ (if $p = 0$, $S_f(B^r) = B^r$). Then there exists a unique $(2p+1)$ -tuple $(\epsilon_1, \dots, \epsilon_p, n_0, \dots, n_p)$, where $\epsilon_j \in \{-1, 1\}$, $n_j \in \mathbb{Z}$ and $n_j \neq 0$ whenever $\epsilon_j = -\epsilon_{j+1}$ so that

$$g = B^{n_0}A^{\epsilon_1}B^{n_1}A^{\epsilon_2} \dots A^{\epsilon_p}B^{n_p} \in \mathbb{F}_2.$$

For example, if $g = A^2$, the associated tuple is $(1, 1, 0, 0, 0)$; and if $g = ABA^{-1}$, the associated tuple is $(1, -1, 0, 1, 0)$.

Let $d_0 = -1$ if $\epsilon_1 = -1$ and $d_0 = 0$ otherwise, $d_p = 1$ if $\epsilon_p = 1$ and $d_p = 0$ otherwise; and let

$$d_j = \frac{\epsilon_j + \epsilon_{j+1}}{2}, \quad 1 \leq j < p.$$

Let now $f : \mathbb{T} \rightarrow \mathbb{T}$ be measurable, and put $c_j(k) = \widehat{f^{d_j}}(k) = \int f(z)^{d_j} z^{-k} d\mu(z)$, where μ is the normalized Haar measure. Let K be the set of $p+1$ -tuples $(k_0, \dots, k_p)$ with $k_j \in \mathbb{Z}$ and such that $k_j = 0$ whenever $d_j = 0$. Set

$$W(k_0, \dots, k_p) = B^{n_0+k_0}A^{\epsilon_1}B^{n_1+k_1}A^{\epsilon_2} \dots A^{\epsilon_p}B^{n_p+k_p}.$$

The map W is injective. Indeed, the only cancellation possible would be if for some j , $\epsilon_j \neq \epsilon_{j+1}$, so that $d_j = 0$. But in that case $k_j = 0$ and so $n_j + k_j = n_j \neq 0$.

Lemma 3. *Let $g \in \mathbb{F}_2$ be a reduced word containing $p \geq 1$ occurrences of $A^{\pm 1}$, $f : \mathbb{T} \rightarrow \mathbb{T}$ measurable, and let $c_j(k)$, W , and K be as above. Then:*

- If h is not in the range of the map W , then $\langle h, S_f(g) \rangle = 0$.
- If $h = W(k_0, \dots, k_p)$, then $\langle h, S_f(g) \rangle = \prod_{j=0}^p c_j(k_j)$.

Proof. By definition of S_f and d_j ,

$$S_f(g) = B^{n_0} f^{d_0}(B) A^{\epsilon_1} B^{n_1} f^{d_1}(B) \cdots A^{\epsilon_p} B^{n_p} f^{d_p}(B).$$

Substituting $f^{d_j}(B) = \sum c_j(k) B^k$ and expanding writes $S_f(g)$ as the orthogonal L^2 sum of words $W(k_0, \dots, k_p)$ each occurring with the coefficient $\prod_{j=0}^p c_j(k_j)$. $\square$

Lemma 4. *Let $f : \mathbb{T} \rightarrow \mathbb{T}$ as before. Let D be a contraction which is diagonal in the group basis for $L^2(M)$, i.e. $Dg = \lambda_g g$, $|\lambda_g| \leq 1$. Then for all $g, h \in \mathbb{F}_2$, $g \neq h$,*

$$|\langle S_f(g), DS_f(h) \rangle| \leq 2\sqrt{\eta_f} + \eta_f,$$

where $\eta_f = 1 - |\tau(f(B))|^2$.

Proof. Suppose first that h has no occurrences of A , so $h = B^r$. Then $DS_f(h) = \lambda_h h$. If g has at least one occurrence of $A^{\pm 1}$, Lemma 3 implies that $S_f(g) \perp h$ and so the inner product is zero. For the same reason, the inner product is also zero if h has at least one occurrence of $A^{\pm 1}$ and g does not. If neither has any occurrence of $A^{\pm 1}$, then $g \neq h$ implies that $S_f(g) = g \perp \lambda_h h = DS_f(h)$. Thus the only interesting case is when both g and h have at least one occurrence of $A^{\pm 1}$.

If this is the case, then Lemma 3 states that for certain integers p and p' , $n_0, \dots, n_p, n'_0, \dots, n'_{p'}$ signs $\epsilon_1, \dots, \epsilon_p, \epsilon'_1, \dots, \epsilon'_{p'}$, and sets K, K' we have:

$$\begin{aligned} S_f(g) &= \sum_{(k_0, \dots, k_p) \in K} \left(\prod_{j=0}^p c_j(k_j) \right) B^{n_0+k_0} A^{\epsilon_1} B^{n_1+k_1} A^{\epsilon_2} \cdots A^{\epsilon_p} B^{n_p+k_p} \\ S_f(h) &= \sum_{(k'_0, \dots, k'_{p'}) \in K'} \left(\prod_{j=0}^{p'} c'_j(k'_j) \right) B^{n'_0+k'_0} A^{\epsilon'_1} B^{n'_1+k'_1} A^{\epsilon'_2} \cdots A^{\epsilon'_{p'}} B^{n'_{p'}+k'_{p'}}. \end{aligned}$$

Thus $S_f(g)$ and $S_f(h)$ are linear combinations of disjoint sets of words (and thus $\langle S_f(g), DS_f(h) \rangle = 0$ since D is diagonal) unless

$$p = p' \quad \text{and} \quad \epsilon_j = \epsilon'_j \quad (1 \leq j \leq p).$$

We may therefore assume these equalities hold. In particular,

$$d_j = d'_j, \quad c_j = c'_j \quad (0 \leq j \leq p).$$

If $d_j = 0$, then $k_j = k'_j = 0$ by definition of K and K' . Thus a word can occur in both sums only if

$$n_j + k_j = n'_j + k'_j,$$

which implies $n_j = n'_j$. Consequently, the inner product is again zero if $n_j \neq n'_j$ whenever $d_j = 0$. Thus we may further assume that $n_j = n'_j$ whenever $d_j = 0$. Put

$$r_j = n'_j - n_j.$$

Since $g \neq h$ and their sign sequences agree, at least one r_j is nonzero. Let us assume that $r_m \neq 0$ (and thus $d_m \neq 0$).

Expanding the sums in the expressions for $S_f(g)$ and $S_f(h)$, we see that the only terms that contribute to the inner product are ones where $n_j + k_j = n'_j + k'_j$, so that $k'_j = k_j - r_j$.

Thus, noting that all $|\lambda_w| \leq 1$,

$$\begin{aligned} |\langle S_f(g), DS_f(h) \rangle| &\leq \sum_{(k_0, \dots, k_p) \in K} \prod_{j=0}^p |c_j(k_j) c_j(k_j - r_j)| \\ &= \prod_{j=0, j \neq m}^p \sum_{k_j \in \mathbb{Z}} |c_j(k_j) c_j(k_j - r_j)| \cdot \sum_{k_m \in \mathbb{Z}} |c_m(k_m) c_m(k_m - r_m)| \\ &\leq \sum_k |c_m(k) c_m(k - r_m)|, \end{aligned}$$

since by Cauchy-Schwarz, $\sum_{k_j} |c_j(k_j) c_j(k_j - r_j)| \leq \|\{c_j(k_j)\}_{k_j}\|_{\ell^2} \|\{c_j(k_j - r_j)\}_{k_j}\|_{\ell^2} \leq \|f^{d_j}\|_2^2 = 1$.

Since $r_m \neq 0$, $d_m \in \{\pm 1\}$; thus the coefficients $c_m(k)$ are either coefficients of f or $\bar{f}$. Thus, denoting by a either r_m or $-r_m$,

$$|\langle S_f(g), DS_f(h) \rangle| \leq \sum_k |c(k) c(k - a)|,$$

where $c(k) = \int f(z) z^{-k} d\mu(z)$ are the Fourier coefficients of f . Then $\eta_f = \sum_{k \neq 0} |c(k)|^2$. In particular, $|c(k)| \leq \sqrt{\eta_f}$ if $k \neq 0$ (and for any k , $|c(k)| \leq 1$). Thus we can write

$$\begin{aligned} \sum_k |c(k) c(k - a)| &= |c(0) c(-a)| + |c(a) c(0)| + \sum_{k \neq 0, k \neq a} |c(k) c(k - a)| \\ &\leq 2\sqrt{\eta_f} + \left(\sum_{k \neq 0, k \neq a} |c(k)|^2 \right)^{1/2} \left(\sum_{k \neq 0, k \neq a} |c(k - a)|^2 \right)^{1/2} \leq 2\sqrt{\eta_f} + \eta_f, \end{aligned}$$

as claimed. $\square$

The following Lemma describes a set of automorphisms that will be used to satisfy the hypothesis of Proposition 2.

Lemma 5. *Let $f : \mathbb{T} \rightarrow \mathbb{T}$ be as before. Let*

$$\Psi_{f,m} = R_{-m} \circ S_f \circ R_m.$$

Then $\Psi_{f,m}$ is a normal trace-preserving automorphism of M , $\Psi_{f,m}(w) = w$, $\Psi_{f,m}(A) = Af(BA^{-m})$ and $\|\Psi_{f,m}(A) - A\|_2 = \|f - 1\|_{L^2(\mathbb{T})}$. Moreover, $\Psi_{f,m}^{-1} = \Psi_{\bar{f},m}$.

Proof. Since both S_f and R_m preserve w , so does $\Psi_{f,m}$. Furthermore, $S_f(R_m(A)) = S_f(A) = Af(B)$. Thus $R_{-m}(S_f(R_m(A))) = R_{-m}(Af(B)) = Af(BA^{-m})$. Finally, $\|\Psi_{f,m}(A) - A\|_2 = \|Af(BA^{-m}) - A\|_2 = \|f(BA^{-m}) - 1\|_2 = \|f - 1\|_{L^2(\mathbb{T})}$ since BA^{-m} is a Haar unitary. $\square$

3.2. Bass-Serre decomposition of $\mathbb{F}_2$. Let H denote the subgroup generated by A and w . As before, set $N = W^*(A, w) = L(H)$. Note that the automorphism R_m satisfies $R_m(A) = A$ and $R_m(w) = w$, so $R_m(h) = h$ for any $h \in H$.

For an element $g \in \mathbb{F}_2$ let $\ell(g)$ be the smallest number of letters B or B^{-1} required to write g as a word in B, B^{-1} and arbitrary elements of H . In particular, $\ell(g) = 0$ iff $g \in H$.

Let $C = BAB^{-1}$. Then H is the subgroup generated by A and C . Moreover, $\mathbb{F}_2$ is the HNN-extension of H : it can be viewed as freely generated by letters A, C, B subject to the relation $BAB^{-1} = C$. The associated Bass-Serre tree $\mathcal{T}$ has as vertex set $\mathbb{F}_2/H$, and edges

between xH and xBH for any $x \in \mathbb{F}_2$. Thus paths in the tree correspond to words in B, B^{-1} and arbitrary elements of H . Thus the geodesic distance $d_{\mathcal{T}}$ on $\mathcal{T}$ satisfies

$$d_{\mathcal{T}}(xH, yH) = \ell(x^{-1}y).$$

For two probability measures μ_1, μ_2 on $\mathbb{F}_2$ define their convolution $\mu_1 * \mu_2$ by

$$\mu_1 * \mu_2(g) = \sum_{g=g_1 g_2} \mu_1(g_1) \mu_2(g_2)$$

and total variation distance

$$\|\mu_1 - \mu_2\|_{TV} = \sup_{E \subset \mathbb{F}_2} |\mu_1(E) - \mu_2(E)| = \frac{1}{2} \sum_{g \in \mathbb{F}_2} |\mu_1(g) - \mu_2(g)|.$$

Note that if μ_1 and μ_2 are probability measures (thus with total mass 1), then

$$\begin{aligned} \|\mu_1 - \mu_2\|_{TV} &= \frac{1}{2} \sum_{g \in \mathbb{F}_2} |\mu_1(g) - \mu_2(g)| \\ &= \frac{1}{2} \sum_{g \in \mathbb{F}_2} (\mu_1(g) + \mu_2(g) - 2 \min(\mu_1(g), \mu_2(g))) \\ &= 1 - \sum_{g \in \mathbb{F}_2} \min(\mu_1(g), \mu_2(g)). \end{aligned}$$

Given $g \in \mathbb{F}_2$, $f : \mathbb{T} \rightarrow \mathbb{T}$ measurable, and $m \in \mathbb{Z}$, define the probability measure $\hat{\mu}_g^{f,m}$ on $\mathbb{F}_2$ by

$$\hat{\mu}_g^{f,m}(h) = |\langle h, \Psi_{f,m}(g) \rangle|^2.$$

The following lemma shows that if m is large enough, then the measures $\hat{\mu}_g^{f,m}$ are, up to a factor that only depends on g and on how close f to 1, given by a convolution.

Lemma 6. *Let $g \in \mathbb{F}_2$ and choose a shortest relative expression for g corresponding to a geodesic from gH to the identity element H in $\mathcal{T}$:*

$$g = h_0 B^{\epsilon_1} h_1 B^{\epsilon_2} \cdots B^{\epsilon_l} h_l, \quad \epsilon_j \in \{\pm 1\}, \quad h_j \in H, \quad l = \ell(g).$$

Let

$$\hat{\nu}_g^{f,m} = \delta_{h_0} * \hat{\mu}_{B^{\epsilon_1}}^{f,m} * \delta_{h_1} * \cdots * \hat{\mu}_{B^{\epsilon_l}}^{f,m} * \delta_{h_l}$$

(with the convention that $\hat{\nu}_g^{f,m} = \delta_g$ if $g \in H$). Then there are constants C_g and m_g so that for any $m > m_g$ and every measurable $f : \mathbb{T} \rightarrow \mathbb{T}$,

$$\|\hat{\nu}_g^{f,m} - \hat{\mu}_g^{f,m}\|_{TV} \leq C_g \eta_f,$$

with $\eta_f = 1 - |\tau(f(B))|^2$.

Proof. Recall that $\Psi_{f,m} = R_{-m} \circ S_f \circ R_m$. Let

$$\mu_g^{f,m}(h) = |\langle h, S_f R_m(g) \rangle|^2$$

and

$$\nu_g^{f,m} = \delta_{h_0} * \mu_{B^{\epsilon_1}}^{f,m} * \delta_{h_1} * \cdots * \mu_{B^{\epsilon_l}}^{f,m} * \delta_{h_l}$$

Since R_{-m} is a group automorphism that fixes H pointwise, it preserves convolutions, total variation distance, and measures δ_{h_j} , and thus

$$\mu_g^{f,m} = (R_m)_* \hat{\mu}_g^{f,m}, \quad \nu_g^{f,m} = (R_m)_* \hat{\nu}_g^{f,m}.$$

Thus the claimed inequality is equivalent to proving that for some C_g and m_g , and arbitrary f and $m > m_g$,

$$\|\nu_g^{f,m} - \mu_g^{f,m}\|_{TV} \leq C_g \eta_f.$$

We first consider the case that $l = \ell(g) > 0$.

Recall that by assumption g has a minimal representation

$$g = h_0 B^{\epsilon_1} h_1 B^{\epsilon_2} \cdots B^{\epsilon_l} h_l, \quad \epsilon_j \in \{\pm 1\}, \quad h_j \in H, \quad l = \ell(g).$$

If $\epsilon_j \neq \epsilon_{j+1}$, then minimality implies that h_j is not a power of A (if $\epsilon_j = 1$) or not a power of BAB^{-1} (otherwise).

Since R_m fixes H pointwise, $R_m(g)$ is obtained from g by replacing each B by BA^m and B^{-1} by $A^{-m}B^{-1}$. The resulting word need not be reduced: it has the form

$$k_0 A^{\epsilon_1 m} k_1 A^{\epsilon_2 m} \cdots A^{\epsilon_l m} k_l$$

where for $0 < j < l$ the words k_j are given by:

$$k_j = \begin{cases} h_j B, & \epsilon_j = \epsilon_{j+1} = 1 \\ h_j, & \epsilon_j = 1, \epsilon_{j+1} = -1 \\ B^{-1} h_j B, & \epsilon_j = -1, \epsilon_{j+1} = 1 \\ B^{-1} h_j & \text{otherwise.} \end{cases}$$

and

$$k_0 = \begin{cases} h_0 B, & \epsilon_1 = 1 \\ h_0 & \text{otherwise;} \end{cases} \quad k_l = \begin{cases} h_l, & \epsilon_l = 1 \\ B^{-1} h_l & \text{otherwise.} \end{cases}$$

Thus cancellation is possible between the start of k_j or the end of k_{j-1} and $A^{\epsilon_j m}$ (note that if $0 < j < l$, k_j must contain at least one letter B , i.e., it has the form $k_j = A^{a_j} v_j A^{b_j}$ with v_j nonempty and starting and ending with a power of B ; and $k_0 = v_0 A^{b_0}$, $k_l = A^{a_l} v_l$, where v_0 may be empty or must end with $B^{\pm 1}$, and v_l must be empty or must start with $B^{\pm 1}$). If we set

$$K = 1 + 2l + \sum_{j=0}^l |h_j|$$

(where we put $|\cdot|$ for the word metric on $\mathbb{F}_2$ associated to the generating set A, B), then there cannot be more than K cancellations between k_j or k_{j-1} and $A^{\epsilon_j m}$. Thus if we choose $m > 2K + 1$, we can ensure that after reduction,

$$R_m(g) = v_0 A^{t_1} v_1 \cdots A^{t_l} v_l$$

where

$$t_j = b_{j-1} + \epsilon_j m + a_j.$$

and same $v_0, \dots, v_l$ as before. Note that the sign of t_j is ϵ_j .

Write each block A^{t_j} as $A^{\epsilon_j K + b_{j-1}} \cdot A^{\epsilon_j(m-2K)} \cdot A^{\epsilon_j K + a_j} = S_j M_j E_j$. Then

$$R_m(g) = v_0 S_1 M_1 E_1 v_1 S_2 M_2 E_2 \cdots v_l$$

is a reduced word.

Let $r = m - 2K$. For each multi-index $\mathbf{k}_j = (k_{j,1}, \dots, k_{j,r-1})$ put

$$M_j(\mathbf{k}_j) = A^{\epsilon_j} B^{k_{j,1}} A^{\epsilon_j} B^{k_{j,2}} \cdots B^{k_{j,r-1}} A^{\epsilon_j}$$

and define

$$V(\mathbf{k}) = v_0 S_1 M_1(\mathbf{k}_1) E_1 \cdots v_l.$$

Define also

$$U_j(\mathbf{k}_j) = \begin{cases} B A^K M_j(\mathbf{k}_j) A^K & \epsilon_j = 1 \\ A^{-K} M_j(\mathbf{k}_j) A^{-K} B^{-1} & \text{otherwise.} \end{cases}$$

Then for any $\mathbf{k}$,

$$V(\mathbf{k}) = h_0 U_1(\mathbf{k}_1) h_1 \cdots U_l(\mathbf{k}_l) h_l.$$

Indeed, in the process of reducing $h_0 U_1(\mathbf{k}_1) h_1 \cdots U_l(\mathbf{k}_l) h_l$ we can perform the same sequence of cancellations as when reducing $h_0 R_m(B^{\epsilon_1}) h_1 \cdots R_m(B^{\epsilon_l}) h_l$, and none of these cancellations occur inside the words $M_j(\mathbf{k}_j)$.

For each $\mathbf{k}$ let

$$F(\mathbf{k}) = \prod_{j=1}^l \prod_{t=1}^{r-1} |c_{\epsilon_j}(k_{j,t})|^2$$

where as before $c_\epsilon(k)$ is the k -th Fourier coefficient of f^ϵ .

Let p be the total number of occurrences of $A^{\pm 1}$ in $V(\mathbf{k})$, and let $d_0, \dots, d_p$ be as described in the paragraph before Lemma 3. Note that if the i -th occurrence of $A^{\pm 1}$ happens inside M_j and not at the end of M_j , then $d_i = \epsilon_j \neq 0$. Thus if we denote by N the total number of nonzero d_j , then $N = l(r-1) + N_0$.

If we put $S = \sum_{j=0}^l |h_j|$, then $p \leq lm + S$ and therefore $N \leq lm + S + 1$. Thus

$$N_0 = N - l(r-1) \leq lm + S + 1 - l(r-1) = S + 1 + (2K+1)l.$$

Lemma 3 then implies that

$$\mu_g^{f,m}(V(\mathbf{k})) = F(\mathbf{k}) |c(0)|^{2N_0}$$

Similarly,

$$\mu_{B^{\epsilon_j}}^{f,m}(U_j(\mathbf{k}_j)) = \left(\prod_{t=1}^{r-1} |c_{\epsilon_j}(k_{j,t})|^2 \right) \cdot |c(0)|^{2(2K+1)} \geq 0$$

and so by definition of convolution and since $V(\mathbf{k}) = h_0 U_1(\mathbf{k}_1) h_1 \cdots U_l(\mathbf{k}_l) h_l$, $\nu_g^{f,m}(V(\mathbf{k}))$ is a sum of non-negative terms, at least one of which is

$$\prod_{j=1}^l \mu_{B^{\epsilon_j}}^{f,m}(U_j(\mathbf{k}_j)) = F(\mathbf{k}) |c(0)|^{2N'_0}, \quad N'_0 = (2K+1)l.$$

Thus

$$\nu_g^{f,m}(V(\mathbf{k})) \geq F(\mathbf{k}) |c(0)|^{2N'_0}.$$

In particular, for any $\mathbf{k}$,

$$\min(\nu_g^{f,m}(V(\mathbf{k})), \mu_g^{f,m}(V(\mathbf{k}))) \geq |c(0)|^{2\max(N_0, N'_0)} F(\mathbf{k}).$$

Since $N_0 \leq S + 1 + (2K+1)l$ and $N'_0 = (2K+1)l$, $\max(N_0, N'_0) \leq S + 1 + (2K+1)l$.

Let $c(0) = \hat{f}(0)$. Because the map $\mathbf{k} \rightarrow V(\mathbf{k})$ is injective,

$$\begin{aligned}
\sum_{h \in \mathbb{F}_2} \min(\nu_g^{f,m}(h), \mu_g^{f,m}(h)) &\geq \sum_{\mathbf{k}} \min(\nu_g^{f,m}(V(\mathbf{k})), \mu_g^{f,m}(V(\mathbf{k}))) \\
&\geq \sum_{\mathbf{k}} F(k) |c(0)|^{2 \max(N_0, N'_0)} \\
&= |c(0)|^{2 \max(N_0, N'_0)} \prod_{j=1}^l \prod_{t=1}^{r-1} \sum_{k_{j,t}} |c_{\epsilon_j}(k_{j,t})|^2 \\
&= |c(0)|^{2 \max(N_0, N'_0)} \prod_{j=1}^l \prod_{t=1}^{r-1} \|f^{\epsilon_j}\|_{L^2}^2 = |c(0)|^{2 \max(N_0, N'_0)}.
\end{aligned}$$

$$\begin{aligned}
\|\hat{\nu}_g^{f,m} - \hat{\mu}_g^{f,m}\|_{TV} &\leq 1 - |c(0)|^{2 \max(N_0, N'_0)} \\
&\leq (1 + |c(0)|^2 + \dots + |c(0)|^{2 \max(N_0, N'_0) - 2})(1 - |c(0)|^2) \\
&\leq (S + 1 + (2K + 1)l)\eta_f = C_g \eta_f,
\end{aligned}$$

with the constant C_g depending only on g .

Let us now consider the case $\ell(g) = 0$, i.e., $g \in H$. Then $\nu_g^{f,m} = \delta_g$. Since $g \in H$, g can be written as a product of $C = BAB^{-1}$ and A . Thus

$$g = B^{n_0} A^{\epsilon_1} B^{n_1} A^{\epsilon_2} \dots A^{\epsilon_p} B^{n_p}$$

In this case 3 shows that $\mu_g^{f,m}(g) \geq |c(0)|^{2r_g}$ where r_g is the total number of letters A in the expansion above. Thus $\|\mu_g^{f,m} - \delta_g\|_{TV} \leq 1 - |c(0)|^{2r_g} \leq r_g \eta_f$. $\square$

3.3. The Operators T_q and Quadratic Forms Q_q . For each $0 \leq q \leq 1$, let

$$T_q : L^2(M) \rightarrow L^2(M)$$

be the diagonal operator

$$T_q \left(\sum \alpha_g g \right) = \sum \alpha_g q^{\ell(g)} g,$$

Then T_q is a positive contraction. For $q = 0$, $T_0 = E_N$, and as $q \rightarrow 1$, $T_q \rightarrow \text{Id}$ strongly. The associated quadratic form $Q_k(x) = \langle T_q x, x \rangle$ is given by

$$Q_q \left(\sum \alpha_g g \right) = \sum |\alpha_g|^2 q^{\ell(g)}.$$

with the convention that $0^0 = 1$.

Note that since R_m fixes H pointwise and replaces each $B^{\pm 1}$ by some word containing only one $B^{\pm 1}$, $\ell(R_m(g)) \leq \ell(g)$. Applying the same logic to $R_{-m} = R_m^{-1}$ shows that $\ell(R_m(g)) = \ell(g)$. In particular, this means that T_q commutes with the unitary implementing R_m on $L^2(M)$.

Corollary 2. *For any $g \in \mathbb{F}_2$, there are constants C_g, m_g so that whenever $f : \mathbb{T} \rightarrow \mathbb{T}$ is measurable, $\eta_f = 1 - |\tau(f(B))|^2$, for any $m > m_g$ and any $0 \leq q \leq 1$,*

$$Q_q(\Psi_{f,m}^{-1}(g)) \geq Q_q(\Psi_{f,m}^{-1}(B))^{\ell(g)} - C_g \eta_f.$$

Proof. By definition,

$$Q_q(\Psi_{f,m}^{-1}(g)) = \sum_h q^{\ell(h)} \hat{\mu}_g^{\bar{f},m} = \sum_h q^{\ell(h)} \mu_g^{\bar{f},m}(h).$$

Since $q \in [0, 1]$, the Total Variation estimate from Lemma 6 implies that

$$\begin{aligned} Q_q(\Psi_{f,m}^{-1}(g)) &\geq \sum_h q^{\ell(h)} \hat{\nu}_g^{\bar{f},m}(h) - C_g \eta_f \geq \prod_{j=1}^{\ell(g)} Q_q(\Psi_{f,m}^{-1}(B^{\epsilon_j})) - C_g \eta_f \\ &\geq Q_q(\Psi_{f,m}^{-1}(B))^\ell - C_g \eta_f, \end{aligned}$$

where the last inequality comes from the definition of $\nu_g^{f,m}(h)$ and the observation that if $h_j \in H$, then $\ell(h_0 s_1 h_1 \dots s_{\ell(g)} h_{\ell(g)}) \leq \sum_{j=1}^{\ell(g)} \ell(s_j)$ $\square$

3.4. Choosing f and m . For $t > 0$ and $m \geq 1$ let

$$f_{m,t}(z) = \exp \left[i \sqrt{\frac{t}{2m}} (z + z^{-1}) \right], \quad \theta_{m,t} = \Psi_{f_{m,t},m} \in \text{Aut}(M).$$

Since $|\exp(iz) - 1| \leq |z|$, we have that

$$|f_{m,t}(z) - 1| \leq \sqrt{\frac{t}{m}} \cdot \left| \frac{z + z^{-1}}{\sqrt{2}} \right|.$$

Thus

$$\begin{aligned} \|f_{m,t} - 1\|_{L^2(\mathbb{T})} &= \left[\int |f_{m,t}(z) - 1|^2 d\mu(z) \right]^{1/2} \\ &\leq \sqrt{t/m} \left[\int \left| \frac{z + z^{-1}}{\sqrt{2}} \right|^2 d\mu(z) \right]^{1/2} = \sqrt{t/m} \left[\int 2 \cos^2 2\pi(t) dt \right]^{1/2} = \sqrt{t/m}, \end{aligned}$$

In particular,

$$\eta_{f_{m,t}} \leq \|f_{m,t} - 1\|^2 \leq t/m.$$

We have made this concrete choice of $f_{m,t}$ for definiteness. In fact, all we need are the following properties of $f_{m,t}$:

Lemma 7. (i) $f_{m,t} : \mathbb{T} \rightarrow \mathbb{T}$ is measurable. (ii) Let $c_m(k) = \int_{\mathbb{T}} \bar{f}_{m,t}(z) z^{-k} d\mu(z) = \widehat{\bar{f}_{m,t}}(k)$, and let $\alpha_m = \left| \widehat{\bar{f}_{m,t}}(0) \right|^2$, $\beta_m = \left| \widehat{\bar{f}_{m,t}}(\pm 1) \right|^2$. Then $\alpha_m = 1 - (t/m) + O(m^{-2})$ and $\beta_m = (t/2m) + O(m^{-2})$. (iii) The Fourier series for f is absolutely convergent.

Proof. Taylor expansion gives $\alpha_m = 1 - (t/m) + O(m^{-2})$ and $\beta_m = (t/2m) + O(m^{-2})$. $\square$

Lemma 8. The automorphism $\theta_{m,t}$ satisfies: $\theta_{m,t}(w) = w$ and $\|\theta_{m,t}(A) - A\|_2 \leq \sqrt{t/m}$.

Proof. Both follow immediately from Lemma 5. $\square$

For $t > 0$ and $q \in [0, 1]$ let

$$F_t(q) = e^{t(q-1)/2} \left(\frac{1 - e^{-t}}{2} + \frac{1 + e^{-t}}{2} q \right).$$

The function $F_t(q)$ will be later shown to be the limit of certain generating functions used to describe the combinatorics of $\theta_{m,t}^{-1}(B)$.

Lemma 9. *For every $t > 0$ and every $\epsilon > 0$ there exists an m_0 so that for all $m > m_0$ and all $0 \leq q \leq 1$,*

$$Q_q(\theta_{m,t}^{-1}(B)) \geq F_t(q) - \epsilon.$$

Proof. Let $c_m(k) = \int_{\mathbb{T}} \bar{f}_{m,t}(z) z^{-k} d\mu(z) = \widehat{\bar{f}_{m,t}}(k)$, and let $\alpha_m = \left| \widehat{\bar{f}_{m,t}}(0) \right|^2$, $\beta_m = \left| \widehat{\bar{f}_{m,t}}(\pm 1) \right|^2$ as in Lemma 7; thus $\alpha_m = 1 - (t/m) + O(m^{-2})$ and $\beta_m = (t/2m) + O(m^{-2})$.

Since $\theta_{m,t}^{-1} = R_{-m} \circ S_{\bar{f}_{m,t}} \circ R_m$, it follows from commutation between T_q and R_m that $Q_q(\theta_{m,t}^{-1}(B)) = Q_q(S_{\bar{f}_{m,t}}(R_m(B))) = Q_q(S_{\bar{f}_{m,t}}(BA^m))$.

Lemma 3 then gives us

$$S_{\bar{f}_{m,t}}(BA^m) = \sum_{\mathbf{n} \in \mathbb{Z}^m} \left(\prod_{j=1}^m c_m(n_j) \right) W(\mathbf{n})$$

where $\mathbf{n} \rightarrow W(\mathbf{n})$ is the injective map

$$W(\mathbf{n}) = BAB^{n_1} AB^{n_2} \dots AB^{n_m}.$$

Thus

$$\begin{aligned} Q_q(\theta_{m,t}^{-1}(B)) &= \sum_{\mathbf{n} \in \mathbb{Z}^m} \left(\prod_{j=1}^m |c_m(n_j)|^2 \right) q^{\ell(W(\mathbf{n}))} \\ (3.1) \quad &\geq \sum_{\mathbf{n} \in \{-1,0,1\}^{m-1} \times \{0\}} \left(\prod_{j=1}^m |c_m(n_j)|^2 \right) q^{\ell(W(\mathbf{n}))}. \end{aligned}$$

We will use the smaller sum (keeping only terms with $n_1, \dots, n_{m-1} \in \{-1, 0, 1\}$ and $n_m = 0$) for our lower estimate. Thus from now on we only consider $n_m = 0$ and each $n_1, \dots, n_{m-1} \in \{-1, 0, 1\}$. Our aim is to bound $\ell(W(\mathbf{n}))$.

Recall that H is generated by A and $C = BAB^{-1}$.

Define $P_0 = BA$, $P_j = P_{j-1} B^{n_j} A$, $1 \leq j \leq m-1$. Since $n_m = 0$,

$$W(\mathbf{n}) = P_{m-1}.$$

We recursively factor

$$P_j = V_j B^{e_j}, \quad e_j \in \{0, 1\}$$

as follows. Initially, $P_0 = BA = CB$, thus we set $V_0 = C$ and $e_0 = 1$. Suppose we factored $P_j = V_j B^{e_j}$. Then

$$P_{j+1} = P_j B^{n_{j+1}} A.$$

We then set V_{j+1}, e_{j+1} according to the table:

$$(V_{j+1}, e_{j+1}) = \begin{cases} (V_j B^{-1} A, 0), & e_j + n_{j+1} = -1, \\ (V_j A, 0) & e_j + n_{j+1} = 0, \\ (V_j C, 1) & e_j + n_{j+1} = 1, \\ (V_j BC, 1) & e_j + n_{j+1} = 2. \end{cases}$$

Because $BA = CB$ and $B^2 A = BCB$, $P_{j+1} = V_{j+1} B^{e_{j+1}}$. We claim that then $P_{j+1} = V_{j+1} B^{e_{j+1}}$. Indeed, there are 6 possible steps:

- (Type 1) Suppose that $e_j + n_{j+1} = -1$, so that $e_j = 0$ and $n_{j+1} = -1$. Then $P_j = V_j B^0$, $V_{j+1} = V_j B^{-1} A$, and $P_{j+1} = P_j B^{-1} A = V_{j+1} B^0$.
- Suppose that $e_j + n_{j+1} = 0$. There are two possibilities:
 - (Type 2) If $e_j = 0$ and $n_{j+1} = 0$, then $P_j = V_j B^0$, $V_{j+1} = V_j A$, and $P_{j+1} = P_j A = V_{j+1} B^0$.
 - (Type 3) If $e_j = 1$ and $n_{j+1} = -1$, then $P_j = V_j B$, $V_{j+1} = V_j A$, and $P_{j+1} = P_j B^{-1} A = V_j A = V_{j+1} B^0$.
- Suppose that $e_j + n_{j+1} = 1$. There are two possibilities:
 - (Type 4) If $e_j = 0$ and $n_{j+1} = 1$, then $P_j = V_j B^0$, $V_{j+1} = V_j C = V_j B A B^{-1}$, and $P_{j+1} = P_j B A = V_{j+1} B$.
 - (Type 5) If $e_j = 1$ and $n_{j+1} = 0$, then $P_j = V_j B$, $V_{j+1} = V_j C = V_j B A B^{-1}$, and $P_{j+1} = P_j A = V_{j+1} B$.
- (Type 6) Suppose finally that $e_j + n_{j+1} = 2$, so that $e_j = 1$ and $n_{j+1} = 1$. Then $P_j = V_j B$, $V_{j+1} = V_j B C = V_j B^2 A B^{-1}$, and $P_{j+1} = P_j B A = V_{j+1} B$.

Let us say that the factorization $P_j = V_j B^{e_j}$ is even (respectively, odd) if $e_j = 0$ (resp., $e_j = 1$).

If the transition from $P_j = V_j B^{e_j}$ to $P_{j+1} = V_{j+1} B^{e_{j+1}}$ is of type 2 or 5, then the parity is unchanged, and $\ell(V_{j+1}) = \ell(V_j)$ since V_j is multiplied by an element of H . We call these transitions **type A**. In this case, $n_{j+1} = 0$.

If the transition is of type 3 or 4, then the parity is reversed, but $\ell(V_{j+1}) = \ell(V_j)$ since V_j is multiplied by an element of H . We call these transitions **type B**. In this case, $|n_{j+1}| = 1$.

Finally, if the transition is of type 1 or 6, then the parity is unchanged, but V_j is changed by the multiplication with either B or B^{-1} as well as an element of H . Thus the length of V_{j+1} could have increased by 1. We call these transitions **type C**. In this case, $|n_{j+1}| = 1$.

Let a_j , b_j , and c_j be the total number of type A, type B, and type C transitions that occurred when moving from $P_0 = V_0 B$ to $P_j = V_j B^{e_j}$. Then

$$\ell(V_j) \leq c_j, \quad \ell(P_j) \leq c_j + e_j.$$

Let $a = a_{m-1}$, $b = b_{m-1}$, $c = c_{m-1}$ be the total number of transitions of each type. Then $a + b + c = m - 1$. Since only type B transitions affect parity, and the initial parity is odd, the final parity is odd iff b is even.

$$\ell(W(\mathbf{n})) = \ell(P_{m-1}) \leq c + 1_{b \text{ even}}.$$

Moreover, the number of indices j for which $|n_j| = 1$ is precisely $c + b$ and the number of j for which $n_j = 0$ is precisely a . Therefore, the term in (3.1) associated to this particular $\mathbf{n}$ is

$$(3.2) \quad \alpha_m^{a+1} \beta_m^{b+c} q^{\ell(W(\mathbf{n}))} \geq \alpha_m^{a+1} \beta_m^{b+c} q^{c+1_{b \text{ even}}}.$$

Note also that if we are given the sequence of transition types, then we can recover $\mathbf{n}$ fully. Indeed, the value of n_{j+1} is completely determined if we know the transition type and the parity e_j : for transitions of type A, $n_{j+1} = 0$; for type B, n_{j+1} is negative iff $e_j = 1$, and for type C, n_{j+1} is positive iff $e_j = 1$. Since $e_0 = 1$, it follows that the sequence of transition types completely determines $\mathbf{n}$. Conversely, knowing $\mathbf{n}$ produces a sequence of $m - 1$ transition types. Therefore, there is a bijection between sequences of $m - 1$ transition types and tuples $\mathbf{n}$. One can realize this transition explicitly using a generating function.

Consider the polynomial $\mathcal{P}_m(X, Y)$ given by

$$\mathcal{P}_m(X, Y) = (\alpha_m + \beta_m X + \beta_m Y)^{m-1} \alpha_m.$$

If one expands the right hand side of this equation, then each term is a product of m terms, each chosen from among α_m , $\beta_m X$ and $\beta_m Y$. Keeping the order of terms in this product, we see that the products are in bijection with sequences of $m - 1$ of three types, with the three terms corresponding to types A, B, and C. Moreover, b and c correspond precisely to the total power of X and Y in the corresponding term.

Let $p_{b,c}^{(m)}$ be the coefficient of $X^b Y^c$ in $\mathcal{P}_m(X, Y)$. Then

$$p_{b,c}^{(m)} = \sum_{\mathbf{n} \text{ has } b \text{ type B and } c \text{ type C}} \alpha_m^{a+1} \beta_m^{b+c},$$

where $a = m - b - c - 1$. Therefore, (3.1) combined with (3.2) implies that

$$Q_q(\theta_{m,t}^{-1}(B)) \geq q \sum_{b,c \geq 0 \text{ with } b \text{ even}} p_{b,c}^{(m)} q^c + \sum_{b,c \geq 0 \text{ with } b \text{ odd}} p_{b,c}^{(m)} q^c.$$

These sums can be recovered by considering the even and odd parts of $\mathcal{P}$ and substituting $X = q$ and $Y = \pm 1$:

$$\sum_{b,c \text{ with } b \text{ even}} p_{b,c}^{(m)} q^c = \frac{1}{2}(\mathcal{P}_m(1, q) + \mathcal{P}_m(-1, q))$$

and

$$\sum_{b,c \text{ with } b \text{ odd}} p_{b,c}^{(m)} q^c = \frac{1}{2}(\mathcal{P}_m(1, q) - \mathcal{P}_m(-1, q)).$$

Substituting this in gives us the final estimate

$$Q_q(\theta_{m,t}^{-1}(B)) \geq \frac{q+1}{2} \mathcal{P}_m(1, q) + \frac{q-1}{2} \mathcal{P}_m(-1, q).$$

Let $\sigma \in \{-1, 1\}$. We now substitute $\alpha_m = 1 - (t/m) + O(m^{-2})$ and $\beta_m = (t/2m) + O(m^{-2})$ into the equation for $\mathcal{P}_m$:

$$\mathcal{P}_m(q, \sigma) = (\alpha_m + \beta_m q + \beta_m \sigma)^{m-1} \alpha_m.$$

For m large enough $\alpha_m + \beta_m(q + \sigma) = 1 - (t/m) + (t/2m)(q + \sigma) + O(m^{-2})$ (where $O(m^{-2})$ is uniform in q). In particular, for large enough m this is positive for all $q \in [-1, 1]$ and both choices of σ . Taking logarithms of both sides gives us

$$\begin{aligned} \log \mathcal{P}_m(q, \sigma) &= (m-1) \log(1 + \frac{t}{2m}(q + \sigma - 2) + O(m^{-2})) + \log(1 - (t/m) + O(m^{-2})) \\ &= (m-1) \frac{t}{2m}(q + \sigma - 2) + O(m^{-1}) \end{aligned}$$

uniformly in q (and pointwise in t). Thus

$$\mathcal{P}_m(q, \sigma) \rightarrow \exp\left(\frac{t}{2}(q + \sigma - 2)\right).$$

It follows that

$$\frac{q+1}{2} \mathcal{P}_m(q, 1) + \frac{q-1}{2} \mathcal{P}_m(q, -1) \rightarrow e^{t(q-1)/2} \left(\frac{1-e^{-t}}{2} + \frac{1+e^{-t}}{2} q \right) = F_t(q).$$

Thus given $\epsilon > 0$ we can choose m_0 so that for any $m > m_0$, for all $q \in [0, 1]$,

$$Q_q(\theta_{m,t}^{-1}(B)) \geq F_t(q) - \epsilon,$$

as claimed. $\square$

3.5. Verification of the hypothesis of Proposition 2.

Lemma 10. *For every $t > 0$, every finite tuple $x_1, \dots, x_n \in L^2(M)$, and every $\varepsilon, \eta > 0$, there exists an m_0 so that for all $m > m_0$*

$$\theta_{m,t}(w) = w, \quad \|\theta_{m,t}(A) - A\|_2 < \eta,$$

and

$$\sum_{i=1}^n \mathcal{Q}_q(\theta_{m,t}^{-1}(x_i)) \geq \sum_{i=1}^n \mathcal{Q}_{F_t(q)}(x_i) - \varepsilon \quad (0 \leq q \leq 1).$$

Proof. Put $\theta_m = \theta_{m,t}^{-1}$. Fix $g \in \mathbb{F}_2$. Then by Corollary 2, there is some m_0 so that for all $m > m_0$

$$Q_q(\theta_m(g)) \geq Q_q(\theta_m(B))^{\ell(g)} - C_g \eta_{f_{m,t}} \geq Q_q(\theta_m(B))^{\ell(g)} - C_g t/m.$$

Thus for any $\epsilon' > 0$ we can use Lemma 9 to deduce that for m large enough,

$$Q_q(\theta_m(g)) \geq F_t(q)^{\ell(g)} - \epsilon', \quad \forall q \in [0, 1].$$

If $g \neq h$, applying Lemma 4 to $R_m(g)$, $R_m(h)$ and $\bar{f}_{m,t}$, and noting that R_{-m} is an isometry and $R_m T_q = T_q R_m$ gives

$$|\langle \theta_m(g), T_q \theta_m(h) \rangle| \leq 2\sqrt{\eta_{\bar{f}}} + \eta_{\bar{f}} \leq 2\sqrt{t/m} + t/m$$

and thus for m large enough,

$$|\langle \theta_m(g), T_q \theta_m(h) \rangle| \leq \epsilon', \quad \forall q \in [0, 1].$$

Now, if $x = \sum \alpha_g g \in L^2(M)$ is a finite sum, choose $\epsilon' = \epsilon \left[\sum_g |\alpha_g| \right]^{-2}$ and apply the definition of Q to deduce

$$\begin{aligned} Q_q(\theta_m(x)) &\geq \left[\sum_g |\alpha_g|^2 F_t(q)^{\ell(g)} \right] - \epsilon' \sum_g |\alpha_g|^2 - \epsilon' \sum_{g \neq h} |\alpha_g| |\alpha_h| \\ &= Q_{F_t(q)}(x) - \epsilon' \left(\sum_g |\alpha_g| \right)^2 = Q_{F_t(q)} - \epsilon. \end{aligned}$$

For a general $x \in L^2(M)$, choose a finite sum x' approximating x , and note that since $T_q \circ \theta_m$ and $T_{F_t(q)}$ are both contractions, $|Q_q(\theta_m(x)) - Q_q(\theta_m(x'))| \leq (\|x\|_2 + \|x'\|_2) \|x - x'\|_2$ similarly for $|Q_{F_t(q)}(x) - Q_{F_t(q)}(x')|$. Choosing an even bigger m_0 ensures that $\|\theta_{m,t}(A) - A\|_2 < \eta$. $\square$

Theorem 3. *Let $\eta > 0$ be given. Let A, B be two free Haar unitaries generating $L(\mathbb{F}_2)$, and let $w = ABA^{-1}B^{-1}$, and let $\mathcal{A} = W^*(w)$. Then there exists a Haar unitary $v \in L(\mathbb{F}_2)$ satisfying $\|v - A\|_2 < \eta$, and so that $L(\mathbb{F}_2) = \mathcal{A} * W^*(v)$.*

Proof. Define recursively $q_0 = 0$, $t_j = (1 - q_j)^2/4$, $q_{j+1} = F_{t_j}(q_j)$. Let $\mathbf{T}_k = T_{q_k}$, $\mathbf{Q}_k = Q_{q_k}$. We claim that with this choice of $\mathbf{T}_k$, the hypothesis of Proposition 2 is then satisfied.

We first show that q_j increase to 1. Indeed, $F_0(q) = q$, $\partial_t F_t(q) \Big|_{t=0} = (1 - q)^2/2$ and $|\partial_t^2 F_t(q)| \leq 2$ for all $t \geq 0$ and $0 \leq q \leq 1$. Thus by Taylor's theorem,

$$F_t(q) \geq q + \frac{t}{2}(1 - q)^2 - t^2.$$

Substituting $t_j = (1 - q_j)^2/4$ gives

$$(3.3) \quad q_{j+1} - q_j \geq \frac{(1 - q_j)^4}{16}.$$

Since $F_t(q) \leq 1$, the sequence q_j must have a limit $a \leq 1$. But if $a < 1$, (3.3) gives that $q_{j+1} - q_j \geq (1 - a)^4/16 > 0$, which is a contradiction. Thus q_j increase to 1.

Thus $\mathbf{T}_k \rightarrow \text{Id}$ strongly; $\mathbf{T}_0 = E_N$ by definition, and each $\mathbf{T}_k$ is a positive contraction.

Fix $k \geq 0$. Given $x_1, \dots, x_n \in L^2(M)$, $\eta > 0$, $\epsilon > 0$ Lemma 10 gives an automorphism $\theta = \theta_{m, t_k}$ that satisfies $\theta(w) = w$, $\|\theta(A) - A\|_2 < \eta$, and

$$\begin{aligned} \sum_{i=1}^n \mathbf{Q}_k(\theta^{-1}(x_i)) &= \sum_{i=1}^n Q_{q_k}(\theta^{-1}(x_i)) \\ &\geq \sum_{i=1}^n Q_{F_{t_k}(q_k)}(x_i) - \epsilon = \sum_{i=1}^n \mathbf{Q}_{k+1}(x_i) - \epsilon \end{aligned}$$

as claimed. $\square$

Corollary 3. *With the notation of the previous Theorem, let $\mathcal{A} = W^*(w) \subset L(\mathbb{F}_2)$. Let (P, τ_P) be any tracial von Neumann algebra, and assume that $\mathcal{A} \subset P$ as a unital von Neumann subalgebra so that the restriction of the group trace on $L(\mathbb{F}_2)$ to $\mathcal{A}$ equals the restriction of τ_P . Then*

$$(L(\mathbb{F}_2), E_{\mathcal{A}}^{L(\mathbb{F}_2)}) *_{\mathcal{A}} (P, E_{\mathcal{A}}^P) \cong L(\mathbb{Z}) * (P, \tau_P),$$

where $E_{\mathcal{A}}^{L(\mathbb{F}_2)}$ and $E_{\mathcal{A}}^P$ denote the trace-preserving conditional expectations.

Proof. Let v be such that $L(\mathbb{F}_2) = W^*(v) * W^*(w)$, and put $\mathcal{B} = W^*(v) \cong L(\mathbb{Z})$. Then

$$L(\mathbb{F}_2) *_{\mathcal{A}} P = (\mathcal{B} * \mathcal{A}) *_{\mathcal{A}} P = \mathcal{B} * P \cong L(\mathbb{Z}) * P,$$

as claimed. $\square$

4. FURTHER GROUP VON NEUMANN ALGEBRA CONSEQUENCES.

In addition to the consequences for finitely generated discrete non-elementary subgroups of $PSL_2(\mathbb{R})$ that were already pointed out, Theorem 3 and Corollary 3 have several other applications.

4.1. Commutator MASA in $L(\mathbb{F}_{2m})$.

Proposition 3. *Let $a_1, v_1, \dots, a_n, v_n$ be generators of $\mathbb{F}_n$ and let $c_j = a_j v_j a_j^{-1} b_j^{-1}$, $w = c_1 \cdots c_n$. Then there exists a Haar unitary v so that*

$$L(\mathbb{F}_{2n}) = W^*(w) * W^*(v, a_2, b_2, \dots, a_n, b_n) \cong W^*(w) * L(\mathbb{F}_{2n-1}).$$

Thus the product-of-commutators MASA in $L(\mathbb{F}_{2m})$ is also freely complemented.

Proof. Let $v \in W^*(a_1, b_1)$ be such that $W^*(a_1, b_1) = W^*(c_1, v)$. Then $c_1, v, a_2, b_2, \dots, a_n, b_n$ are freely independent generators of $L(\mathbb{F}_2)$. Let $r = c_2 \cdots c_n \in \langle a_2, b_2, \dots, a_n, b_n \rangle$. Then $c_1 r, v, a_2, b_2, \dots, a_n, b_n$ are again freely independent generators. Since $w = c_1 r$, the claimed isomorphism follows. $\square$

With the notation of the previous Proposition, let $\mathcal{A} = W^*(w) \subset L(\mathbb{F}_{2n})$. Let (P, τ_P) be any tracial von Neumann algebra, and assume that $\mathcal{A} \subset P$ and $\tau_{L(\mathbb{F}_{2n})}|_{\mathcal{A}} = \tau_P|_{\mathcal{A}}$. Then

$$(L(\mathbb{F}_{2n}), E_{\mathcal{A}}^{L(\mathbb{F}_{2n})}) *_{\mathcal{A}} (P, E_{\mathcal{A}}^P) \cong L(\mathbb{F}_{2n-1}) * (P, \tau_P).$$

In particular, if Λ is any group $x \in \Lambda$ an element of infinite order. Then $L(\Lambda *_{x=w^{-1}} \mathbb{F}_{2m}) \cong L(\Lambda) * L(\mathbb{F}_{2m-1})$. Thus, using [6], we obtain:

Proposition 4. *Let Λ be any amenable group, and let $x \in \Lambda$ be an element of infinite order. Then $L(\Lambda *_{x^{-1}=\prod_{j=1}^n a_j b_j a_j^{-1} b_j^{-1}} \langle a_1, b_1, \dots, a_n, b_n \rangle) \cong L(\mathbb{F}_{2n})$.*

4.2. Non-orientable surfaces of negative curvature. If $\mathcal{N}_h$ is a closed non-orientable surface of genus $h \geq 3$, then its fundametal group is given by the presentation

$$\pi_1(\mathcal{N}_h) = \langle a, b, x_1, \dots, x_{h-2} \mid aba^{-1}b^{-1}x_1^2 \cdots x_{h-2}^2 = 1 \rangle.$$

Thus $\pi_1(\mathcal{N}_h) = \langle a, b \rangle *_{aba^{-1}b^{-1}=(x_1^2 \cdots x_{h-2}^2)^{-1}} \langle x_1, \dots, x_{h-2} \rangle$. Since $x_1^2 \cdots x_{h-2}^2$ has infinite order in $\mathbb{F}_{h-2}$ we can use Corollary 3 to deduce that $L(\mathcal{N}_h) \cong L(\mathbb{F}_{h-1})$. Combined with Theorem 1 this gives:

Theorem 4. *Let S be a closed connected surface of negative Euler characteristic. Then $L(\pi_1(S)) \cong L(\mathbb{F}_{1-\chi(S)})$.*

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